

# JEFFIN SOLOMON ASIR J S

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## Education

College of Engineering, Guindy

B.E Computer Science and Engineering; **CGPA: 8.51**

Bell Matric Higher Secondary School

HSC; *Percentage: 98.00*

Chennai, India

**August 2023 – Expected June 2027**

Tirunelveli, India

**June 2021 – April 2023**

## Skills Summary

**Languages:** C, Java, Python, JavaScript, Bash

**Databases:** MongoDB, Oracle SQL, MySQL

**Developer Tools:** Git, Postman

**Relevant Coursework:** Data Structures, Algorithms, Operating Systems, Computer Networks, DBMS, OOP

**Frameworks:** React, Node.js, Express.js

**Systems:** Linux, Docker, REST APIs

**Networking:** WebSockets, TCP/IP

## Technical Experience

Summer Intern at DCSE | College of Engineering, Guindy

June 2025 – July 2025

**Lab Evaluation System** | <https://github.com/simple-sid/lab-eval-cn>

**Tech Stack:** React, Node.js, MongoDB, Docker, WebSockets, Monaco, Xterm.js, Bash, Python, tcpdump

- **Engineered** a Docker-based lab evaluation platform giving each student an isolated Linux container, a self-hosted **Monaco** browser IDE supporting fully offline operation in restricted exam environments, and a live SSH terminal over **Xterm.js** and **WebSockets**.
- **Developed** an automated evaluation engine in **Bash** and **Python** that captures live loopback traffic with **tcpdump** and parses raw packets to validate real TCP/UDP protocol behavior, rather than trusting self-reported console output.
- **Implemented** self-healing container lifecycle management with automatic port-conflict recovery, slot-based module expiry, multi-file autosave, and instructor dashboards for live session, performance, and container monitoring.
- **Deployed** department-wide across the **5th Semester Computer Networks Lab (240+ students)**, following successful beta testing with **80+ students**.

## Projects

**Explainable Insider Threat Detection** | <https://github.com/jeffin-asir/xai-masquerader-detector>

**Tech Stack:** Python, TensorFlow, Keras, Scikit-learn, SHAP, LIME, NSL-KDD Dataset

- **Developed** an explainable deep-learning model (**DenseAttnDNN**) for insider threat detection, combining IDS signals with engineered behavioral features to achieve **99.28% accuracy** and **1.8%** false positive rate on the **NSL-KDD** benchmark.
- **Benchmarked** against **7** baseline detectors, outperforming all on accuracy, precision, recall, and F1-score (cutting FPR to **1.8%** vs up to **11.4%**); added **SHAP/LIME** explainability with human-in-the-loop feedback for continuous retraining.

**Stock-A-Mock | Mock Stock Market Web App** | <https://github.com/jeffin-asir/Stock-A-Mock>

**Tech Stack:** React, Node.js, Express, OracleSQL, JWT, REST API

- **Implemented** transactional stock trading via **Oracle PL/SQL** stored procedures (`buy_stock`, `sell_stock`) that row-lock accounts with `FOR UPDATE`, atomically updating balance, holdings, and weighted-average cost basis across a normalized schema (Users, DematAccounts, Stocks, Transactions).
- **Developed** PL/SQL functions computing portfolio-level **CAGR** and **XIRR** (via Newton-Raphson root-finding over irregular cash flows), integrated with the **finnhub.io** market API for live price feeds and real-time profit/loss analytics.

## Achievements

**GATE 2026 – Computer Science and Information Technology (CS)**

February 2026

- Secured **AIR 7264** with GATE Score **504 (43.16/100)** among **2,11,020** candidates nationwide; qualified above General cutoff (**30.0**) in CS & IT. Demonstrated strong fundamentals across core subjects including DSA, Operating Systems, DBMS, Computer Networks, and Theory of Computation.

## Certificates

**NPTEL – Database Management System (IIT Kharagpur)**

- **Elite + Top 1%** distinction; scored **76%** across assignments and proctored exams, ranking in the **top 1%** of **8,423** certified candidates; built strong foundations in database design and query optimization.

**Cisco – Network Security (Coursera)**

- Covered threat mitigation (MITM, DoS), secure communication protocols, and access control; gained practical knowledge of network defense strategies; authorized by Cisco Learning and Certifications.